NMCI102 Engineering Mathematics 2 | **Department of Mathematics & Computing**
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Tutorial III

1

1.1 (a)

Given that, $(2x^2 + 4y)\, dx + (4x + y - 1)dy = 0$
Comparing the given equation with $Mdx + Ndy = 0$, we have

$$M(x, y) = 2x^2 + 4y$$
$$N(x, y) = 4x + y - 1$$

Differentiating M and N partially w.r.to y and x respectively, we get

$\frac{\partial M}{\partial y} = 4$ and $\frac{\partial N}{\partial x} = 4$

So clearly, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
Hence, the given equation is exact and its solution is

$\int M(x, y)dx$ [Treating y as constant in $M(x, y)$]+$\int$ [Terms in $N(x, y)$ not including x]$dy = c$

$$\Rightarrow \quad \int \left(2x^2 + 4y\right) dx + \int (y - 1)dy = c$$

$$\Rightarrow \quad \frac{2x^3}{3} + 4xy + \frac{y^2}{2} - y = c \text{ [Using Power rule of integration]}$$

Therefore the required solution is,

$$\frac{2x^3}{3} + 4xy + \frac{y^2}{2} - y = c$$

Note: Power Rule of Integration,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad \text{for} \quad n \neq -1.$$

1.2 (b)

$(1 + 2xy\cos x^2 - 2xy)\,dx + (\sin x^2 - x^2)\,dy = 0$
Comparing above eq. with $Mdx + Ndy = 0$
Here $M = 1 + 2xy\cos x^2 - 2xy$ and $N = \sin x^2 - x^2$

Now differentiating M, N partially w.r.to y and x respectively, we get

$$\frac{\partial M}{\partial y} = 2x\cos x^2 - 2x \text{ and } \frac{\partial N}{\partial x} = 2x\cos x^2 - 2x$$

Clearly, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
Hence eq. is exact and its solution is

$$\int M(x,y)dx \quad [\text{Treating } y \text{ as constant in } M(x,y)] + \int [\text{Terms in } N(x,y) \text{ not including } x]dy = c$$

$$\Rightarrow \quad \int \left(1 + 2xy\cos x^2 - 2xy\right) dx + \int (0)\cdot dy = c$$

$$\Rightarrow \quad x + 2y\sin x^2 - \frac{2yx^2}{2} = c$$

$$\Rightarrow \quad x + 2y\sin x^2 - yx^2 = c$$

1.3 (c)

Given that, $xdx + ydy + \frac{xdy - ydx}{x^2+y^2} = 0$

Rewriting the given equation as,

$$\left(x - \frac{y}{x^2+y^2}\right) dx + \left(y + \frac{x}{x^2+y^2}\right) dy = 0$$

Comparing above eq. with $M(x,y)dx + N(x,y)dy = 0$

Then, $M=x-\frac{y}{x^2+y^2}$ and $N=y+\frac{x}{x^2+y^2}$

Differentiating M and N partially with respect to y and x respectively we get,

$$\frac{\partial M}{\partial y}=0-\frac{1\cdot\left(x^2+y^2\right)-y\cdot 2y}{\left(x^2+y^2\right)^2}=\frac{-\left(x^2-y^2\right)}{\left(x^2+y^2\right)^2}=\frac{y^2-x^2}{\left(x^2+y^2\right)^2}$$

and $\frac{\partial N}{\partial x}=0+\frac{1\left(x^2+y^2\right)-x\cdot 2x}{\left(x^2+y^2\right)^2}=\frac{y^2-x^2}{\left(x^2+y^2\right)^2}$.

Thus $\quad \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$.

Hence eq. is exact, and therefore its solution is

$$\int M(x,y)dx \quad \text{[Treating } y \text{ as constant in } M(x,y)]+\int \text{[Terms in } N(x,y) \text{ not including } x]dy=C$$

$$\Rightarrow \int\left(x-\frac{y}{x^2+y^2}\right)dx+\int y\,dy=C$$

$$\Rightarrow \frac{x^2}{2}-y\cdot\left(\frac{1}{y}\right)\tan^{-1}\left(\frac{x}{y}\right)+\frac{y^2}{2}=C$$

$$\Rightarrow x^2+y^2-2\tan^{-1}(x/y)=c \text{ [Here } 2C=c \text{]}$$

This is the required solution.

1.4 (d)

Given that, $\left(y^2e^{xy^2}+4x^3\right)dx+\left(2xye^{xy^2}-3y^2\right)dy=0$.

Comparing the given eq. with $Mdx+Ndy=0$

here,

$$M=y^2e^{xy^2}+4x^3$$
$$N=2xye^{xy^2}-3y^2$$

Differentiating M and N partially w.r.to y and x respectively we get,

$$\frac{\partial M}{\partial y}=2ye^{xy^2}+y^2\cdot 2xye^{xy^2}$$

$$\text{and } \frac{\partial N}{\partial x}=2ye^{xy^2}+2xye^{xy^2}\cdot y^2$$

Note that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
Hence, the given eq. is exact and its solution is

$$\int M(x,y)dx \quad \text{[Treating } y \text{ as constant in } M(x,y)]+\int \text{[Terms in } N(x,y) \text{ not including } x]dy = c$$

$$\Rightarrow \int \left(y^2e^{xy^2} + 4x^3\right) dx + \int \left(-3y^2\right) dy = c$$
$$\Rightarrow y^2 \cdot \frac{1}{y^2}e^{xy^2} + 4\left(\frac{x^4}{4}\right) - 3 \cdot \frac{y^3}{3} = c$$
$$\Rightarrow e^{xy^2} + x^4 - y^3 = c$$

This is the required solution.

1.5 (e)

Given that, $\left[y\left(1+\frac{1}{x}\right) + \cos y\right] dx + [x + \log x - x \sin y]dy = 0$

Comparing the given eq. with $Mdx + Ndy = 0$

$$M = y\left(1+\frac{1}{x}\right) + \cos y$$
$$N = x + \log x - x\sin y$$

Differentiating M and N partially w.r.to y and x respectively, we get

$$\frac{\partial M}{\partial y} = 1 + \frac{1}{x} - \sin y = \frac{\partial N}{\partial x}$$

Hence, eq. is exact and so its solution is

$$\int M(x,y)dx \quad \text{[Treating } y \text{ as constant in } M(x,y)]+\int \text{[Terms in } N(x,y) \text{ not including } x]dy = c$$

$$\Rightarrow \int \left(y + \frac{y}{x} + \cos y\right) dx + \int 0dy = c$$
$$\Rightarrow yx + y\log x + x\cos y = c \text{This is the required solution.}$$

2

2.1 (a)

Given that,

$$(2x^2 + y^2 + x)\,dx + xy\,dy = 0 \tag{1}$$

Compare (1) with $M(x,y)\,dx + N(x,y)\,dy = 0$, we get

$$M(x,y) = 2x^2 + y^2 + x$$
$$N(x,y) = xy$$

So, Differentiating M and N partially w.r.to y and x respectively, we get

$$\frac{\partial M}{\partial y} = 2y$$
$$\frac{\partial N}{\partial x} = y$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So the given differential equation is not exact.

Now,

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{2y}{xy} - \frac{y}{xy} = \frac{1}{x} = f(x)$$
$$\Rightarrow \frac{1}{xy}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{2y}{xy} - \frac{y}{xy} = \frac{1}{x} = f(x)$$

So integrating factor (I.F.) = $e^{\int f(x)\,dx} = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x$

Now multiplying both side of (1) by I.F. = x

$$(2x^3 + y^2x + x^2)\,dx + x^2y\,dy = 0$$

Let,

$$M_1(x,y) = (2x^3 + y^2x + x^2) \text{ and } N_1(x,y) = x^2y$$

Then,

$$\frac{\partial M_1}{\partial y} = 2xy = \frac{\partial N_1}{\partial x}.$$

Hence, the differential equation is exact now.

Now,

$$\int M_1(x,y)dx + \int [N_1(x,y)\,dy] = C$$

$$\int M_1(x,y)dx \quad \text{[Treating } y \text{ as constant in } M_1(x,y)]+\int \text{[Terms in } N_1(x,y) \text{ not including } x]dy = C$$

$$\Rightarrow \int \left(2x^3 + y^2x + x^2\right) dx + \int 0\,dy = C$$

$$\Rightarrow \frac{2x^4}{4} + \frac{y^2x^2}{2} + \frac{x^3}{3} = C.$$

[Using power rule of integration]

$$\Rightarrow 3x^4 + 3x^2y^2 + 2x^3 = c \text{ , where } c = 6C$$

Therefore, the required solution is,

$$3x^4 + 3x^2y^2 + 2x^3 = c.$$

Note: Power Rule of Integration,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad \text{for} \quad n \neq -1.$$

2.2 (b)

Given that,

$$x^2\frac{dy}{dx} + xy = \sqrt{1 - x^2y^2}$$

$$
\begin{aligned}
&\Rightarrow x\frac{dy}{dx} + y = \sqrt{\frac{1-x^2y^2}{x^2}} \\
\text{Let, } xy = v &\Rightarrow x\frac{dy}{dx} + y = \frac{dv}{dx} \text{[Differentiating both sides]} \\
&\Rightarrow \frac{dv}{dx} = \sqrt{\frac{1-v^2}{x^2}} \text{ [Putting the value } v = xy \text{]} \\
&\Rightarrow x\,dv - \sqrt{1-v^2}\,dx = 0 \\
&\Rightarrow \sqrt{1-v^2}\,dx - x\,dv = 0 \qquad (1)
\end{aligned}
$$

[As in the above equation we can separate the variables so we can evaluate the solution directly]

That is,

$$
\begin{aligned}
\frac{dx}{x} - \frac{dv}{\sqrt{1-v^2}} &= 0 \\
\Rightarrow \frac{dx}{x} &= \frac{dv}{\sqrt{1-v^2}} \\
\Rightarrow \log x &= \sin^{-1} v + \text{C}
\end{aligned}
$$

Where C is constant.

Putting the value of $v = xy$ we get,

$$\log x = \sin^{-1} xy + \text{C}$$

$$\Rightarrow \ \sin^{-1} xy = \log x + \text{c} \text{ , where } c = -C$$

This is the required solution.

2.3 (c)

$$\text{we have, } (3xy^2 - y^3)\,dx - (2x^2y - xy^2)\,dy = 0$$

Here, $M(x, y) = 3x^2y - y^3$ and $N(x, y) = -(2x^2y - xy^2)$

$$\begin{aligned}
\text{I.F.} &= \frac{1}{Mx+Ny} \quad \text{(Since it's a homogeneous type)}\\
&= \frac{1}{3x^2y^2 - xy^3 - 2x^2y^2 + xy^3}\\
&= \frac{1}{x^2y^2}
\end{aligned}$$

Then, $IF = \frac{1}{x^4y^2}$

Now multiply I.F. to the given ODE we get:

$$\left(\frac{3}{x} - \frac{y}{x^2}\right) dx + \left(\frac{1}{x} - \frac{2}{y}\right) dy = 0$$

So here, $M_1(x,y) = \frac{3}{x} - \frac{y}{x^2}$ and $N_1(x,y) = \frac{1}{2y}$.

Clearly, $\frac{\partial M_1}{\partial y} = -\frac{1}{x^2}$ and $\frac{\partial N_1}{\partial x} = -\frac{1}{x^2}$. Hence, the differential equation is in exact form.

Now,

$$\int M_1(x,y)dx \quad \text{[Treating } y \text{ as constant in } M_1(x,y)] + \int \text{[Terms in } N_1(x,y) \text{ not including } x]dy = C$$

$$\Rightarrow \int \left(\frac{3}{x} - \frac{y}{x^2}\right) dx - \int \frac{1}{2y} dy = C$$

$$\Rightarrow 3\log x + \frac{y}{x} - 2\log y = C, \quad \text{where } C \text{ is a constant.}$$

Therefore, the required solution is:

$$3\log x + \frac{y}{x} - 2\log y = C.$$

Note: Here we use the following integration formulas:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad \text{for } n \neq -1,$$
$$= \log x + C \quad \text{for } n = -1.$$

2.4 (d)

we have,

$$(y + xy^2)\, dx + (x - x^2 y)\, dy = 0$$

$$\text{Here, } M(x,y) = y + xy^2 \text{ and } N(x,y) = x - x^2 y$$

Now,

$$\frac{1}{Mx - Ny} = \frac{1}{xy + x^2y^2 - xy + x^2y^2}$$
$$= \frac{1}{2x^2y^2}$$

$$\text{So, I.F. is } \frac{1}{2x^2y^2}$$

Multiplying the given equation by the I.F., we get:

$$\frac{1}{2x^2y^2}(y + xy^2)\, dx + \frac{1}{2x^2y^2}(x - x^2y)\, dy = 0$$
$$\Rightarrow \left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx + \left(\frac{1}{2xy^2} - \frac{1}{2y}\right) dy = 0$$

Consider, $M_1(x,y) = \frac{1}{2x^2y} + \frac{1}{2x}$ and $N_1(x,y) = \frac{1}{2xy^2} - \frac{1}{2y}$.

Then,

$$\frac{\partial M_1}{\partial y} = -\frac{1}{2x^2y^2}$$

and

$$\frac{\partial N_1}{\partial x} = -\frac{1}{2x^2y^2}$$

Hence the differential equation is exact.

Now,

$$\int M_1(x,y)dx \quad \text{[Treating } y \text{ as constant in } M_1(x,y)]+\int \text{[Terms in } N_1(x,y) \text{ not including } x]dy = C$$

$$\Rightarrow \int \left(\frac{1}{2x^2y}+\frac{1}{2x}\right)dx+\int -\frac{1}{2y}dy = C$$
$$\Rightarrow -\frac{1}{2xy}+\frac{1}{2}\log x-\frac{1}{2}\log y = C$$
$$\Rightarrow -\frac{1}{2xy}+\frac{1}{2}\log(\frac{x}{y}) = C$$
$$\Rightarrow -\frac{1}{xy}+\log(\frac{x}{y}) = c \text{ , where } c = 2C$$

Therefore, the required solution is:

$$-\frac{1}{xy}+\log(\frac{x}{y}) = c$$

2.5 (e)

Given that,

$$(x^2+y^2+x)\,dx+xy\,dy = 0 \tag{1}$$

$$\text{Now, } \frac{\partial M}{\partial y} = 2y$$
$$\frac{\partial N}{\partial x} = y$$

$$\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right) = \frac{1}{xy}(2y-y) = \frac{1}{x} = f(x)\text{(say)} \tag{2}$$

$$\begin{aligned}\text{Then I.F.} &= e^{\int f(x)dx}\\ &= e^{\int \frac{1}{x}dx}\\ &= e^{\ln x}\\ &= x\end{aligned}$$

Multiply this I.F. on both sides of the given Differential equation

$$(x^3 + xy^2 + x^2)\, dx + x^2 y\, dy = 0 \tag{3}$$

Let us consider,

$$M_1(x, y) = x^3 + xy^2 + x^2$$

and

$$N_1(x, y) = x^2 y.$$

Then,

$$\frac{\partial M_1}{\partial y} = 2xy = \frac{\partial N_1}{\partial x}.$$

Hence, the differential equation (3) is exact.

Now,

$\int M_1(x, y) dx$ [Treating y as constant in $M_1(x, y)$]+$\int$ [Terms in $N_1(x, y)$ not including x]$dy = c$

$$\Rightarrow \int \left(x^3 + xy^2 + x^2\right) dx + \int 0 dy = c.$$

$$\Rightarrow \frac{x^4}{4} + \frac{x^2 y^2}{2} + \frac{x^3}{3} = c.$$

This is the required solution.

3

(a) Given,

$$\frac{dy}{dx} + \frac{x}{1 - x^2} y = x\sqrt{y} \tag{4}$$

It is a Bernoullis equation; we have to convert it into a linear differential equation.
Now dividing both sides by $\sqrt{y}$ in equation (1) we get

$$\frac{1}{\sqrt{y}}\frac{dy}{dx}+\frac{x}{1-x^2}\sqrt{y}=x \tag{5}$$

Let

$$\sqrt{y}=u$$

Differentiating w.r.t x we get

$$\implies \frac{du}{dx}=\frac{1}{2\sqrt{y}}\frac{dy}{dx}$$

$$\implies 2\frac{du}{dx}=\frac{1}{\sqrt{y}}\frac{dy}{dx}$$

Putting this in equation (2), we get

$$2\frac{du}{dx}+\frac{x}{1-x^2}u=x$$

$$\implies \frac{du}{dx}+\frac{x}{1-x^2}\frac{u}{2}=\frac{x}{2}$$

It is now in the form of linear differential equation: $\frac{du}{dx}+P(x)u=Q(x)$
Here, $P(x)=\frac{x}{2(1-x^2)}\ and\ Q(x)=\frac{x}{2}$
Now,

$$\begin{aligned}
(Integrating\ Factor) I.F =& e^{\int Pdx}\\
=& e^{\int \frac{x}{2(1-x^2)}dx} \quad \text{[Let } 1-x^2=t \implies dt=-2xdx \implies \frac{-dt}{2}=xdx\text{]}\\
=& e^{-\frac{1}{4}\int\frac{dt}{t}}\\
=& e^{-\frac{1}{4}lnt}\\
=& \frac{1}{t^4}\\
\implies I.F = & \frac{1}{(1-x^2)^{\frac{1}{4}}}
\end{aligned}$$

Then,

$$\begin{aligned}
u.(I.F) &= \int (I.F)Q(x)dx + c \\
\frac{u}{(1-x^2)^{\frac{1}{4}}} &= \int \frac{x}{2(1-x^2)^{\frac{1}{4}}}dx + c \\
&= \frac{1}{2}\int \frac{x}{(1-x^2)^{\frac{1}{4}}}dx + c \;\; \text{[Let } 1-x^2 = t \implies dt = -2xdx \implies \frac{-dt}{2} = xdx\text{]} \\
&= -\frac{1}{4}\int \frac{dt}{t^{\frac{1}{4}}} + c \\
&= -\frac{t^{(1-\frac{1}{4})}}{(1-\frac{1}{4})} + c \\
&= -\frac{t^{\frac{3}{4}}}{3} + c \\
&= -\frac{(1-x^2)^{\frac{3}{4}}}{3} + c
\end{aligned}$$

So, the required solution,

$$\begin{aligned}
u &= \frac{-1}{3}(1-x^2) + c(1-x^2)^{\frac{1}{4}} \\
\implies \sqrt{y} &= \frac{-1}{3}(1-x^2) + c(1-x^2)^{\frac{1}{4}}
\end{aligned}$$

(b) Given

$$\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y \quad ...(1)$$

Now dividing both sides by cos^2y in equation (1), we get

$$\Rightarrow \frac{1}{\cos^2 y}\frac{dy}{dx} + x\frac{\sin 2y}{\cos^2 y} = x^3$$

$$\Rightarrow \sec^2 y\frac{dy}{dx} + x\frac{2\sin y\cos y}{\cos^2 y} = x^3$$

$$\Rightarrow \sec^2 y\frac{dy}{dx} + 2x\tan y = x^3 \quad ...(2)$$

Let

$$\tan y = u$$

Differentiating w.r.t x we get

$$\Rightarrow \frac{du}{dx} = \sec^2 y \frac{dy}{dx}$$

Now putting this in equation (2), we get

$$\frac{du}{dx} + 2xu = x^3$$

It is now in the form of a linear differential equation:

$$\frac{du}{dx} + P(x)u = Q(x)$$

Now Integrating Factor

$$I.F = e^{\int 2x\,dx} = e^{x^2}$$

Now,

$$u.e^{x^2} = \int x^3 e^{x^2}\,dx + c$$

$$= \int x^3 e^{x^2} dx + c$$

$$= \int x x^2 e^{x^2} dx + c$$

[Let $x^2 = t \implies dt = 2xdx \implies \frac{dt}{2} = xdx$]

$$= \frac{1}{2} \int te^t dt + c$$

$$= \frac{1}{2} e^t (t-1) + c$$

$$= \frac{1}{2} e^{x^2} (x^2 - 1) + c$$

$$\Rightarrow e^{x^2} tany = \frac{1}{2} e^{x^2} (x^2 - 1) + c$$

(c) Given equation is

$$x\frac{dy}{dx} + y\log y = xye^x \tag{1}$$

Now dividing both sides by xy in equation (1), we get

$$\frac{1}{y}\frac{dy}{dx} + \frac{\log y}{x} = e^x \tag{2}$$

Let

$$\log y = u$$

Differentiating w.r.t x we get,

$$\Rightarrow \frac{du}{dx} = \frac{1}{y}\frac{dy}{dx}$$

From (2)

$$\frac{du}{dx} + \frac{u}{x} = e^x$$

It is in the form of linear differential equation.

$$I.F = e^{\int \frac{1}{x}dx} = e^{\log x} = x$$

Now the required solution

$$u.x = \int e^x.x\,dx + c$$

$$\text{or} \quad u.x = xe^x - e^x + c$$

$$\text{or} \quad x.\log y = e^x(x-1) + c$$

(d) Given,

$$\frac{dy}{dx} + 2xy = xy^3 \tag{1}$$

Dividing by y^3 in both sides, we get,

$$\frac{1}{y^3}\frac{dy}{dx} + \frac{2x}{y^2} = x \tag{2}$$

Let
$$\frac{1}{y^2} = u$$
Differentiating w.r.t x we get,
$$\Rightarrow \frac{du}{dx} = -\frac{2}{y^3}\frac{dy}{dx}$$
$$\Rightarrow \frac{1}{y^3}\frac{dy}{dx} = -\frac{1}{2}\frac{du}{dx}$$

Putting this value in equation (2) we get

$$-\frac{1}{2}\frac{du}{dx} + 2xu = x$$

$$\Rightarrow \frac{du}{dx} - 4xu = -2x$$

It is in the form of a linear differential equation.
Integrating Factor (I.F)= $e^{\int -4x\,dx} = e^{-2x^2}$

Now the required solution
$$u.e^{-2x^2} = \int -2xe^{-2x^2}\,dx + c \tag{3}$$

Let $-2x^2 = t \Rightarrow -4x\,dx = dt \Rightarrow -2x\,dx = \frac{dt}{2}$

So from (3),
$$u.e^{-2x^2} = \int \frac{e^t}{2}\,dt + c$$
$$= \frac{1}{2}e^t + c$$
$$= \frac{1}{2}e^{-2x^2} + c$$

$$\Rightarrow \frac{1}{y^2} = \frac{1}{2} + ce^{2x^2}$$

(e) Given,
$$\frac{dy}{dx}\sin x - y\cos x + y^2 = 0$$

$$\Rightarrow \sin x\, dy - y\cos x\, dx + y^2\, dx = 0$$

$$\Rightarrow \frac{\sin x\, dy - y\cos x\, dx}{y^2} + dx = 0$$

[Multiply both sides by (-1)]

$$\Rightarrow \frac{y\cos x\, dx - \sin x\, dy}{y^2} - dx = 0$$

Now integrating both sides, we get

$$\Rightarrow \int d\left(\frac{\sin x}{y}\right) - \int dx = 0$$

$$\Rightarrow \frac{\sin x}{y} - x = c, c\, is\; arbitrary\; constant$$

$$\Rightarrow \sin x = y(x + c)$$

(f) Given,

$$(1 + y^2) + (x - e^{-\tan^{-1} y})\frac{dy}{dx} = 0$$

$$or\;\; (x - e^{-\tan^{-1} y})\frac{dy}{dx} = -(1 + y^2)$$

$$or\;\; \frac{dx}{dy} = \frac{e^{-\tan^{-1} y} - x}{1 + y^2}$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1 + y^2} = \frac{e^{-\tan^{-1} y}}{1 + y^2}$$

It is in the form of LDE

Now, Integrating Factor $= e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}$

Then the required solution

$$xe^{\tan^{-1} y} = \int e^{\tan^{-1} y} \cdot \frac{e^{-\tan^{-1} y}}{1+y^2}\, dy + c$$

$$\Rightarrow xe^{\tan^{-1} y} = \int \frac{dy}{1+y^2} + c$$

$$\Rightarrow xe^{\tan^{-1} y} = \tan^{-1} y + c$$

$$\Rightarrow x = (c + \tan^{-1} y)e^{-\tan^{-1} y}$$

(g) Given,

$$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2} \tag{1}$$

Dividing (1) by y^2 in both sides, we get,

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x^2} \tag{2}$$

Let

$$\frac{1}{y} = u$$

Differentiating w.r.t x we get,

$$\Rightarrow \frac{du}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$$

$$\Rightarrow -\frac{du}{dx} = \frac{1}{y^2}\frac{dy}{dx}$$

From equation (2)

$$-\frac{du}{dx} + \frac{u}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{du}{dx} - \frac{u}{x} = -\frac{1}{x^2}$$

It is in the form of a linear differential equation.
Integrating Factor (I.F) $= e^{\int -\frac{1}{x}dx} = e^{-\log x} = \frac{1}{x}$

Now the required solution,

$$u.\frac{1}{x} = \int \frac{1}{x}(-\frac{1}{x^2})\,dx + c$$

$$\Rightarrow \frac{u}{x} = -\int \frac{1}{x^3}\,dx + c$$

$$\Rightarrow \frac{u}{x} = \frac{1}{2x^2} + c$$

$$\Rightarrow u = \frac{1}{2x} + cx$$

$$\Rightarrow \frac{1}{y} = \frac{1}{2x} + cx$$

$$\Rightarrow x = \frac{y}{2} + \frac{c}{2}x^2y$$

(h) Given,

$$\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2} \tag{1}$$

Dividing (1) by e^y in both sides, we get,

$$\Rightarrow e^{-y}\frac{dy}{dx} + e^{-y}\frac{1}{x} = \frac{1}{x^2}$$

Let

$$e^{-y} = u$$

Differentiating w.r.t x we get,

$$\Rightarrow \frac{du}{dx} = -e^{-y}\frac{dy}{dx}$$

$$\Rightarrow -\frac{du}{dx} + \frac{u}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{du}{dx} - \frac{u}{x} = -\frac{1}{x^2}$$

It is in the form of a linear differential equation.
Integrating Factor $= e^{-\int \frac{1}{x}dx} = e^{-\log x} = \frac{1}{x}$

Now the required solution

$$u.\frac{1}{x} = -\int \frac{1}{x^3}\,dx + c = \frac{1}{2x^2} + c$$
$$\implies e^{-y}x = \frac{1}{2x^2} + c$$
$$\implies 2x = e^{y}(1 + cx^2)$$

(i) Given,

$$\frac{dy}{dx} + \frac{1}{x}\sin 2y = x^2\cos^2 y \tag{1}$$

Dividing (1) by $cos^2 y$ in both sides, we get,

$$\Rightarrow \frac{1}{\cos^2 y}\frac{dy}{dx} + \frac{1}{x}\frac{\sin 2y}{\cos^2 y} = x^3$$

$$\Rightarrow \sec^2 y\frac{dy}{dx} + \frac{1}{x}\frac{2\sin y\cos y}{\cos y} = x^3$$

$$\Rightarrow \sec^2 y\frac{dy}{dx} + \frac{2}{x}\tan y = x^3 \tag{2}$$

Let

$$\tan y = u$$

Differentiating w.r.t x, we get,

$$\Rightarrow \frac{du}{dx} = \sec^2 y\frac{dy}{dx}$$

From equation (2),

$$\frac{du}{dx} + \frac{2u}{x} = x^3$$

it is in the form of a linear differential equation.
Integrating Factor (I.F) $= e^{\int \frac{2}{x}dx} = e^{\log x^2} = x^2$

Now the required solution,

$$
\begin{aligned}
u.x^2 &= \int x^2 x^3 \, dx + c \\
\implies u.x^2 &= \frac{x^6}{6} + c \\
\implies \tan y &= (\frac{x^4}{6} + cx^{-2}) \\
\implies 6x^2 \tan y &= x^6 + c
\end{aligned}
$$

(j) Given,

$$\frac{dy}{dx} + \frac{y}{x}\log y = \frac{y}{x^2}(\log y)^2 \tag{1}$$

Dividing (1) by $y(\log y)^2$ in both sides, we get,

$$\Rightarrow \frac{1}{y(\log y)^2}\frac{dy}{dx} + \frac{1}{x}\log y = \frac{1}{x^2}$$

Let

$$\frac{1}{\log y} = u$$

Differentiating w.r.t x, we get,

$$\Rightarrow \frac{du}{dx} = -\frac{1}{(\log y)^2}\frac{1}{y}\frac{dy}{dx}$$

$$\Rightarrow -\frac{du}{dx} = \frac{1}{y(\log y)^2}\frac{dy}{dx}$$

$$\Rightarrow -\frac{du}{dx} + \frac{u}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{du}{dx} - \frac{u}{x} = -\frac{1}{x^2}$$

$$
\begin{aligned}
\text{Integrating Factor} &= e^{\int -\frac{1}{x} dx} \\
&= \frac{1}{x}
\end{aligned}
$$

Now,

$$u.\frac{1}{x} = \int -\frac{1}{x^3}\,dx + c$$
$$\Rightarrow \frac{u}{x} = (\frac{1}{2x^2} + c)$$
$$\Rightarrow u = \frac{1}{2x} + cx$$
$$\Rightarrow \frac{1}{\log y} = \frac{1}{2x} + cx$$
$$\Rightarrow x = \log y(cx^2 + \frac{1}{2})$$

4

4a: To find: Orthogonal Trajectory for $xy = a^2$

The given family of curves is:

$$xy = a^2$$

Differentiating with respect to x:

$$x\frac{dy}{dx} + y = 0 \implies \frac{dy}{dx} = -\frac{y}{x}$$

The slope of the orthogonal trajectory is the negative reciprocal:

$$\frac{dy}{dx} = \frac{x}{y}$$

Rearranging and integrating:

$$y\,dy = x\,dx$$

$$\int y\,dy = \int x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + C$$

$$y^2 - x^2 = 2C$$

The orthogonal trajectories are a family of hyperbolas represented by:

$$y^2 - x^2 = C'$$

where $C' = 2C$ is an arbitrary constant.

4b. To find: Orthogonal Trajectory for $3xy = x^3 - a^3$

The given family of curves is: $3xy = x^3 - a^3$

First, we differentiate with respect to x:

$$3x\frac{dy}{dx} + 3y = 3x^2$$

$$\frac{dy}{dx} = \frac{x^2 - y}{x}$$

The slope of the orthogonal trajectory is the negative reciprocal, hence the slope of the orthogonal trajectory is given by:

$$\frac{dy}{dx} = \frac{x}{y - x^2}$$

$$(y - x^2)\,dy = x\,dx$$

$$y\,dy - x^3\,dy = x\,dx$$

To solve the differential equation:

$$\frac{dy}{dx} = \frac{x}{y - x^2},$$

$$(y - x^2)\,dy = x\,dx.$$

Let $u = y - x^2$. Then, the derivative of u with respect to x is:

$$\frac{du}{dx} = \frac{dy}{dx} - 2x.$$

From the original equation, we know that $\frac{dy}{dx} = \frac{x}{u}$, so:

$$\frac{du}{dx} = \frac{x}{u} - 2x.$$

$$\frac{du}{dx} = \frac{x - 2xu}{u}.$$

$$\frac{du}{dx} = \frac{x(1 - 2u)}{u}.$$

$$\frac{u}{1 - 2u}\,du = x\,dx.$$

Integrating both sides

$$\int \frac{u}{1 - 2u}\,du = \int x\,dx.$$

For the left-hand side, we use substitution. Let $v = 1 - 2u$, so $dv = -2\,du$, and $du = -\frac{1}{2}\,dv$.

$$\int \frac{u}{v}\left(-\frac{1}{2}\right)\,dv = -\frac{1}{2}\int \frac{u}{v}\,dv.$$

Since $u = \frac{1-v}{2}$, substitute:

$$-\frac{1}{2}\int\frac{\frac{1-v}{2}}{v}\,dv=-\frac{1}{4}\int\frac{1-v}{v}\,dv.$$

solving the differential equation we get: $v = Ce^{v-2x^2}$. By making the back substitutions we get following as the general solution to the differential equation:

$$x^2 = y - \frac{1}{2} + Ce^{-2y}$$

where C is the constant of integration.
4d: To find: Orthogonal trajectory for $r = a(1 - cos\theta)$
Differentiating with respect to θ we get :

$$\frac{dr}{d\theta} = asin(\theta) \tag{6}$$

$$\tag{7}$$

Dividing the equation of curve by the above equation

$$\frac{r}{\frac{dr}{d\theta}} = \frac{1-cos(\theta)}{sin(\theta)} \tag{8}$$

$$\implies r\frac{d\theta}{dr} = \frac{1-cos\theta}{sin\theta} \tag{9}$$

Using the orthogonality condition, $r\frac{d\theta}{dr} = -\frac{1}{r}\frac{dr}{d\theta}$ we get:

$$-\frac{1}{r}\frac{dr}{d\theta} = \frac{1-cos\theta}{sin\theta} \tag{10}$$

Rearranging and integrating both sides we get:

$$logr = logC\frac{sin^2\theta}{1-cos\theta} \tag{11}$$

$$\implies r = C(1+cos\theta) \tag{12}$$

Where C is arbitrary constant.
4e To find: Orthogonal Trajectory for $r^n = a^n cos(n\theta)$

Differentiating both sides with respect to θ we get:

$$nr^{n-1}\frac{dr}{d\theta} = -a^n nsin(n\theta) \tag{13}$$
$$\implies r^{n-1}\frac{dr}{d\theta} = -a^n sin(n\theta) \tag{14}$$

Dividing the equation of the curve by the equation (7) we get:

$$\frac{r}{\frac{dr}{d\theta}} = -cot(n\theta) \tag{15}$$
$$\implies r = -\frac{dr}{d\theta}cot(n\theta) \tag{16}$$

Using the orthogonality condition, $r\frac{d\theta}{dr} = -\frac{1}{r}\frac{dr}{d\theta}$ we get:

$$tan(n\theta) = r\frac{d\theta}{dr} \tag{17}$$
$$\implies \frac{dr}{r} = cot(n\theta)d(\theta) \tag{18}$$

Integrating both sides :

$$log(r) = \frac{1}{n}log(sin(n\theta))C') \tag{19}$$
$$\implies r^n = Csin(n\theta) \tag{20}$$

Where C is arbitary constant.

4f: To find: Orthogonal trajectory for $x^2 - y^2 = a^2$

Differentiating both sides with respect to x we get:

$$2x - 2y\frac{dy}{dx} = 0 \tag{21}$$
$$\implies \frac{dy}{dx} = \frac{x}{y} \tag{22}$$

Using the orthogonality condition: $\frac{dy}{dx} = -\frac{dx}{dy}$ the above equation becomes:

$$\frac{dy}{dx} = -\frac{y}{x}$$

Rearranging the above equation and on integrating we get:

$$logy = -logCx \tag{23}$$
$$\implies xy = C. \tag{24}$$

Where C is an arbitrary constant.

4g: **To find:** Orthogonal Trajectory for $\frac{x^2}{a^2} + \frac{y^2}{a^2+\lambda^2} = 1$

Differentiating both sides with respect to x we get:

$$\frac{2x}{a^2} + \frac{2y}{a^2+\lambda^2}\frac{dy}{dx} = 0 \tag{25}$$
$$\implies \frac{a^2}{a^2+\lambda^2} = -\frac{x}{y}\frac{dy}{dx} \tag{26}$$

The curve equation can be rewritten as:

$$x^2 + \frac{a^2}{a^2+\lambda^2}y^2 = a^2 \tag{27}$$

Substituting the value of $\frac{a^2}{a^2+\lambda^2}$ from equation (7) we get:

$$x^2 - xy\frac{dx}{dy} = a^2 \tag{28}$$

Using the orthogonality condition: $\frac{dy}{dx} = -\frac{dx}{dy}$ the above equation becomes:

$$x^2 + xy\frac{dy}{dx} = a^2 \tag{29}$$
$$\implies x^2dx + xy.dy = a^2dx \tag{30}$$
$$\implies ydy = \frac{a^2 - x^2}{x}dx \tag{31}$$

integrating both sides of equation (12) we get:

$$x^2 + y^2 = 2a^2 logx + C.$$

Where C is an arbitrary constant.

4h: To find: Orthogonal Trajectory for $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

Differentiating both sides with respect to x we get:

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx} = 0 \tag{32}$$

$$\implies \frac{dy}{dx} = -\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}} \tag{33}$$

From Orthogonality condition, $\frac{dy}{dx} = -\frac{dx}{dy}$ the above equation becomes:

$$-\frac{dx}{dy} = -\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}} \tag{34}$$

Integrating both sides of equation (30) we get:

$$x^{\frac{4}{3}} - y^{\frac{4}{3}} = C$$

Where C is an arbitrary constant.

4i: Same as question 4e with n = 2.

4j: To find: Orthogonal Trajectory for $r^n sin(n\theta) = a^n$

The given family of curves is:

$$r^n sin(n\theta) = a^n$$

Taking log both sides

$$nlogr + log(sin(n\theta)) = nloga$$

Differentiating with respect to θ:

$$\frac{d(nlogr)}{d\theta} + \frac{d(nsin(n\theta))}{d\theta} = 0 \implies \frac{n}{r}\frac{dr}{d\theta} + \frac{ncos(n\theta)}{sin(n\theta)} = 0$$

On further simplifying we get:

$$\frac{rd\theta}{dr} = -tan(n\theta)$$

The slope of the orthogonal trajectory is the negative reciprocal, therefore we have

$$\frac{dr}{rd\theta} = tan(n\theta)$$

Rearranging and integrating:

$$\int \frac{1}{r}\,dr = \int tan(n\theta)\,dx$$

$$r = C(secn\theta)^{\frac{1}{n}}$$

$$r^n cos(n\theta) = C^n$$

The orthogonal trajectories are a family of hyperbolas represented by:

$$r^n cos(n\theta) = C^n$$

C is an arbitrary constant.

4k: To find: Orthogonal trajectory for $r = a(1 + cos\theta)$
Differentiating with respect to θ we get :

$$\frac{dr}{d\theta} = -asin(\theta) \tag{35}$$

(36)

Dividing the equation of curve by the above equation

$$\frac{r}{\frac{dr}{d\theta}} = -\frac{1+cos(\theta)}{sin(\theta)} \tag{37}$$
$$\implies r\frac{d\theta}{dr} = \frac{1-cos\theta}{sin\theta} \tag{38}$$

Using the orthogonality condition, $r\frac{d\theta}{dr} = -\frac{1}{r}\frac{dr}{d\theta}$ we get:

$$\frac{1}{r}\frac{dr}{d\theta} = \frac{1+cos\theta}{sin\theta} \tag{39}$$

Rearranging and integrating both sides we get:

$$logr = logC(1-cos\theta) \tag{40}$$
$$\implies r = C(1-cos\theta) \tag{41}$$

Where C is arbitrary constant.

4l: To find: Orthogonal trajectory for $r = \frac{2a}{(1+cos\theta)}$
Differentiating with respect to θ we get :

$$\frac{dr}{d\theta} = \frac{2asin(\theta)}{(1+cos\theta)^2} \tag{42}$$
$$\tag{43}$$

Dividing the equation of curve by the above equation

$$\frac{r}{\frac{dr}{d\theta}} = \frac{1+cos(\theta)}{sin(\theta)} \tag{44}$$
$$\implies r\frac{d\theta}{dr} = \frac{1+cos\theta}{sin\theta} \tag{45}$$

Using the orthogonality condition, $r\frac{d\theta}{dr} = -\frac{1}{r}\frac{dr}{d\theta}$ we get:

$$\frac{1}{r}\frac{dr}{d\theta} = -\frac{1 + cos\theta}{sin\theta} \tag{46}$$

Rearranging and integrating both sides we get:

$$logr = -logC(1 - cos\theta) \tag{47}$$

$$\Longrightarrow r = \frac{C}{(1 - cos\theta)} \tag{48}$$

Where C is arbitrary constant.

5

5.1 a

To show: Family of Parabolas $x^2 = 4a(y + a)$ is self orthogonal. We have

$$\begin{aligned} y &= \frac{x^2}{4a} - a \\ \Rightarrow \frac{dy}{dx} &= \frac{2x}{4a} \\ \Rightarrow a &= \frac{x}{2\frac{dy}{dx}} \end{aligned} \tag{49}$$

Putting this in 49, we have

$$y = \frac{x}{2}\frac{dy}{dx} - \frac{x}{2}\frac{dx}{dy} \tag{50}$$

To obtain the orthogonal family of 49 we replace $\frac{dy}{dx}$ with $\frac{-dx}{dy}$ in 50 we have,

$$y = -\frac{x}{2}\frac{dx}{dy} + \frac{x}{2}\frac{dy}{dx} \tag{51}$$

We note that eq 51 is same as equation 50 So solving 51 we get family 49.

Hence, the family49 is self orthogonal.

5.2 b

Prove that the system of confocal conics $\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}=1, \lambda$ being the parameter is self orthogonal.

$$\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}=1 \tag{1}$$

Now, differentiating w.r.to x, we get

$$\begin{aligned}
\frac{2x}{a^2+\lambda}+\frac{2y}{b^2+\lambda}\frac{dy}{dx}&=0\\
x\left(b^2+\lambda\right)+y\left(a^2+\lambda\right)\frac{dy}{dx}&=0\\
\lambda\left(x+y\frac{dy}{dx}\right)&=-\left(b^2x+a^2y\frac{dy}{dx}\right)\\
\lambda&=\frac{-\left(b^2x+a^2y\frac{dy}{dx}\right)}{x+y\frac{dy}{dx}}
\end{aligned}$$

we have,

$$\begin{aligned}
a^2+\lambda=a^2-\frac{\left(b^2x+a^2y\frac{dy}{dx}\right)}{x+y\frac{dy}{dx}}&=\frac{\left(a^2-b^2\right)x}{x+y\frac{dy}{dx}}\\
b^2+\lambda=b^2-\frac{\left(b^2x+a^2y\frac{dy}{dx}\right)}{x+y\frac{dy}{dx}}&=\frac{-\left(a^2-b^2\right)y\frac{dy}{dx}}{x+y\frac{dy}{dx}}
\end{aligned}$$

putting these in (1)

$$\begin{gathered}
\frac{x^2\left(x+y\frac{dy}{dx}\right)}{\left(a^2-b^2\right)x}-\frac{y^2\left(x+y\frac{dy}{dx}\right)}{\left(a^2-b^2\right)y\frac{dy}{dx}}=1\\
\left(x+y\frac{dy}{dx}\right)\left(x-y\frac{xx}{dy}\right)=\left(a^2-b^2\right)
\end{gathered} \tag{2}$$

Note that: To obtain the orthogonal family of (1), we replace $\frac{dy}{dx}$ with - $\frac{dx}{dy}$ in (2), we get

$$\left(x-y\frac{dx}{dx}\right)\left(x+y\frac{dy}{dx}\right)=a^2-b^2$$

So, solving the above gives us the orthogonal family of (1).

6

Given function $f(x,y) = x\sin y + y\cos x \quad |x| \le a, |y| \le b$

Now,

$$\begin{aligned}
|f(x,y_1) - f(x,y_2)| &= |x\sin y_1 + y_1\cos x - (x\sin y_2 + y_2\cos x)| \\
&= |x(\sin y_1 - \sin y_2) + (y_1 - y_2)\cos x| \\
&\le |x|\,|\sin y_1 - \sin y_2| + |y_1 - y_2|\,|\cos x| \\
\text{Now,} \qquad &\le |x|\,|y_1 - y_2| + |y_1 - y_2| \ \text{ as } \{|\sin x - \sin y| \le |x - y|\} \\
&= (|x| + 1)\,|y_1 - y_2| \\
\Rightarrow \quad |f(x,y_1) - f(x,y_2)| &\le (a+1)\,|y_1 - y_2| \\
&= L\,|y_1 - y_2|
\end{aligned}$$

Thus, $f(x,y)$ satisfies Lipschitz Condition with Lipschitz Constant $L = a + 1$.

7

Given function $f(x,y) = y^{2/3}$ on $R : |x| \le 1, |y| \le 1$. There is no Lipchitz constant in any interval containing zero since

$$\frac{|f(x,y) - f(x,0)|}{|y - 0|} = \frac{1}{|y^{1/3}|} \to \infty \text{ as } y \to 0$$

8

Consider the differential equation:

$$M(x,y)dx + N(x,y)dy = 0. \tag{1}$$

Assume that equation (1) has a general solution $f(x,y) = C$. Then the total differential of $f(x,y)$ is:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0. \tag{2}$$

Since f is the general solution, from equations (1) and (2), we get:

$$\frac{dy}{dx} = -\frac{M}{N} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}.$$

This implies:

$$\frac{\frac{\partial f}{\partial x}}{M} = \frac{\frac{\partial f}{\partial y}}{N}. \tag{3}$$

If we denote the typical ratio in equation (3) by a function $\mu(x,y)$, then:

$$\frac{\partial f}{\partial x} = \mu M, \quad \frac{\partial f}{\partial y} = \mu N.$$

On multiplying equation (2) by $\mu(x,y)$, it becomes:

$$\mu M dx + \mu N dy = 0.$$

This simplifies to:

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0,$$

Which is an exact differential equation.

This argument shows that if equation (2) has a general solution, then it has at least one integrating factor $\mu(x,y)$. It has infinitely many integrating factors. For instance, if $F(f)$ is any function of

$f(x,y)$, then:

$$F(f)df = d[F(f)],$$

And hence:

$$F(f)\mu(Mdx + Ndy) = d[F(f)].$$

Thus, $\mu F(f)$ is also an integrating factor for equation (1) as $d[F(f)]$ is integrable. For different F we can get different integrating factor $\mu F(f)$.

Hence, given differential equations possess infinite solutions.

9

Consider the initial value problem (IVP):

$$y' = f(x,y), \quad y(x_0) = y_0.$$

1. If f is continuous in a region $R = \{(x,y) : |x - x_0| \leq a, |y - y_0| \leq b\}$, and f is bounded in that region ($|f(x,y)| \leq M$), then there exists at least one solution $y(x)$ in the interval:

$$|x - x_0| \leq h,$$

where $h = \min(a, b/M)$.

2. If f satisfies the Lipschitz condition in R:

$$|f(x,y_1) - f(x,y_2)| \leq K|y_1 - y_2|,$$

then the IVP has a unique solution in the interval $|x - x_0| \leq h$.

Application to Given IVP

Consider the IVP:

$$\frac{dy}{dx} = 16 + y^2, \quad y(0) = 0.$$

1. Existence of Solution:

The function $f(x,y) = 16 + y^2$ is continuous in the region $R = \{(x,y) : |x| \leq a, |y| \leq b\}$.

- The bound for $f(x, y)$ is:

$$|f(x, y)| = |16 + y^2| \leq 16 + b^2.$$

Let $M = 16 + b^2$.

By Picard's theorem, a solution exists in the interval:

$$|x - x_0| \leq h,$$

where $h = \min(a, b/M)$. and $x_0 = 0$
Here, a is free but, b/M has a limit(here we will take the maximum value)

To find a specific interval of existence, let:

$$g(b) = \frac{b}{M} = \frac{b}{16 + b^2}.$$

Differentiate to find critical points:

$$g'(b) = \frac{16 - b^2}{(16 + b^2)^2}.$$

Setting $g'(b) = 0$, we get $b = 4$. at $b = 4$, $g(b)$ is maximum one can go for the sign of $g''(b = 4)$, which is negative or check the values of $g(b)$ near $b = 4$.
At $b = 4$, evaluate:

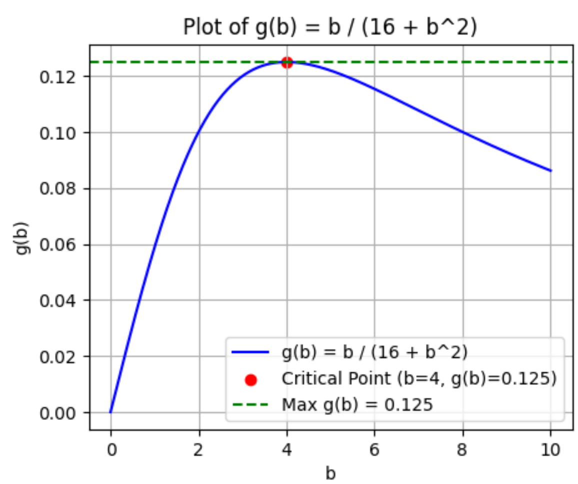

Figure 1: Caption

$$g(4) = \frac{4}{16 + 4^2} = \frac{4}{32} = \frac{1}{8}.$$

Thus, the interval of existence is:

$$|x| \leq h = \frac{1}{8}.$$

Code is shared at https://www.overleaf.com/read/tgwzccwpmqpv#f74c16

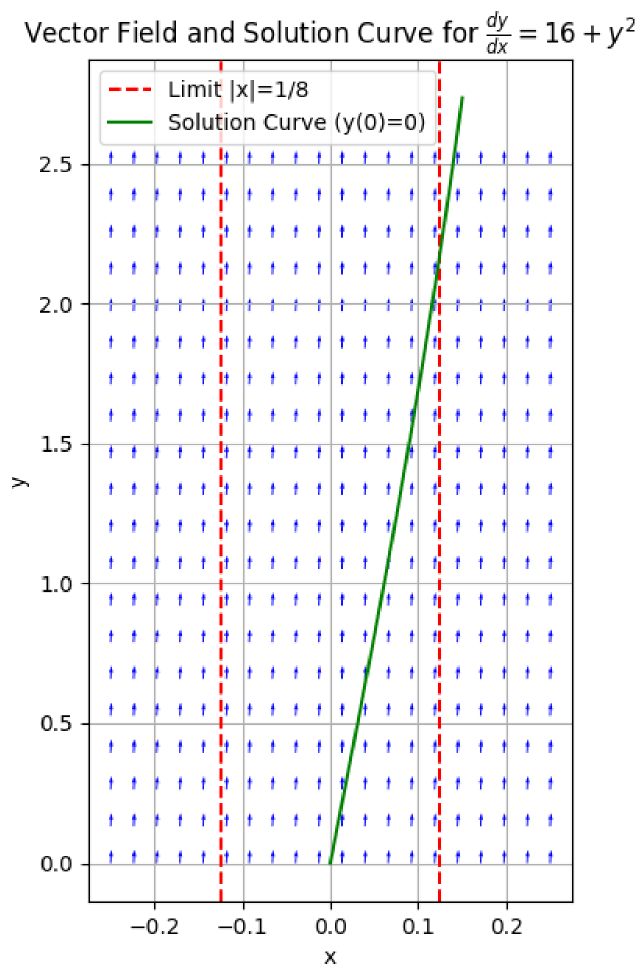

Figure 2: Caption

10

Given that, $\frac{dy}{dx} = 2 - \frac{y}{x}$, $y(1) = 2$

Here, $f(x, y) = 2 - \frac{y}{x}$, $y_0 = 2$, $x_0 = 1$

Picard's method is:

$$y_{n+1} = y_0 + \int_{x_0}^{x} f(x, y_n)\, dx$$

For $n = 0$:

$$\begin{aligned}
y_1 &= y_0 + \int_1^x f(x, y_0)\, dx \\
&= 2 + \int_1^x \left(2 - \frac{2}{x}\right) dx \\
&= 2 + [2x - 2\ln x]_1^x \\
&= 2 + 2x - 2\ln x - 2 + 2\ln 1 \\
&= 2(x - \ln x)
\end{aligned} \tag{52}$$

For $n = 1$:

$$\begin{aligned}
y_2 &= y_0 + \int_1^x f(x, y_1)\, dx \\
&= 2 + \int_1^x \left[2 - \frac{2(x - \ln x)}{x}\right] dx \\
&= 2 + \int_1^x \left[2 - 2 + 2\frac{\ln x}{x}\right] dx \\
&= 2 + 2\int_1^x \ln x\, d\ln x \\
&= 2 + 2(\frac{(\ln x)^2}{2} - 0) \\
&= 2 + (\ln x)^2
\end{aligned} \tag{53}$$

11

11.1 (i)

Given the differential equation:

$$y' = 2xy, \quad y(0) = 1$$

Using the Picard iteration method:

$$f(t, y) = 2ty$$

The formula for the iteration is:

$$y_{n+1} = y_0 + \int_0^x f\big(t, y_n(t)\big)\, dt$$

Iterations:

$$\begin{aligned}
y_1(x) &= 1 + \int_0^x f\big(t, y_0(t)\big)\, dt = 1 + \int_0^x 2t\, dt = 1 + x^2 \\
y_2(x) &= 1 + \int_0^x 2t(1 + t^2)\, dt = 1 + \left[2\frac{t^2}{2} + 2\frac{t^4}{4}\right]_0^x = 1 + x^2 + \frac{x^4}{2} \\
y_3(x) &= 1 + \int_0^x 2t \cdot (1 + t^2 + \frac{t^4}{2})\, dt \\
&= 1 + \int_0^x (2t + 2t^3 + t^5) dt \\
&= 1 + \left[2\frac{t^2}{2} + 2\frac{t^4}{4} + \frac{t^6}{6}\right]_0^x \\
&= 1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6}
\end{aligned} \tag{54}$$

Thus, the solution is:

$$y(x) = 1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \cdots = e^{x^2}$$

11.2 (ii)

Given the differential equation:

$$y' = y - x, \quad y(0) = 2$$

Using the Picard iteration method:

$$f(t, y) = y - t$$

The formula for the iteration is:

$$y_{n+1} = y_0 + \int_0^x f\big(t, y_n(t)\big)\, dt$$

Iterations:

$$\begin{aligned}
y_1(x) &= 2 + \int_0^x (2 - t)\, dt = 2 + 2x - \frac{x^2}{2} \\
y_2(x) &= 2 + \int_0^x \big(2 + t - \frac{t^2}{2}\big)\, dt = 2 + 2x + \frac{x^2}{2} - \frac{x^3}{6} \\
y_3(x) &= 2 + \int_0^x (2 + 2t + \frac{t^2}{2} - \frac{t^3}{6} - t)\, dt \\
&= 2 + \int_0^x \big(2 + t + \frac{t^2}{2} - \frac{t^3}{6}\big)\, dt \\
&= 2 + 2x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24}
\end{aligned}$$

Thus, the solution is:

$$y(x) = x + e^x + 1 = 2 + 2x + \frac{x^2}{2} + \frac{x^3}{6} + ...$$